

**12**  $f$  is inversely proportional to  $d^2$

$$f = 3.5 \text{ when } d = 8$$

(a) Find an equation for  $f$  in terms of  $d$ .

(2)

(b) Find the positive value of  $d$  when  $f = 10$   
Give your answer correct to 3 significant figures.

$$d =$$

(2)

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**(Total for Question 12 is 4 marks)**